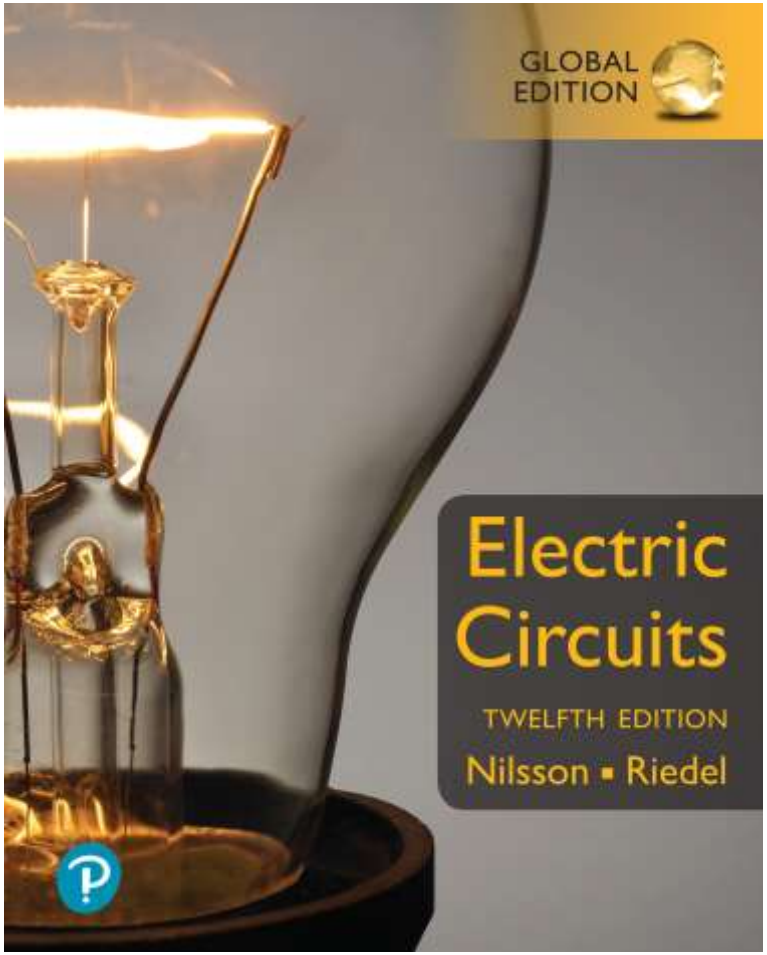
Electric Circuits

Twelfth Edition, Global Edition



Chapter 6 – Part 3

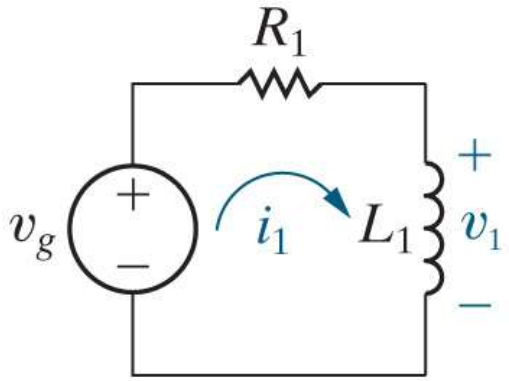
Inductance, Capacitance, and Mutual Inductance

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (1 of 15)

Start with a simple RL circuit with a time-varying voltage source:



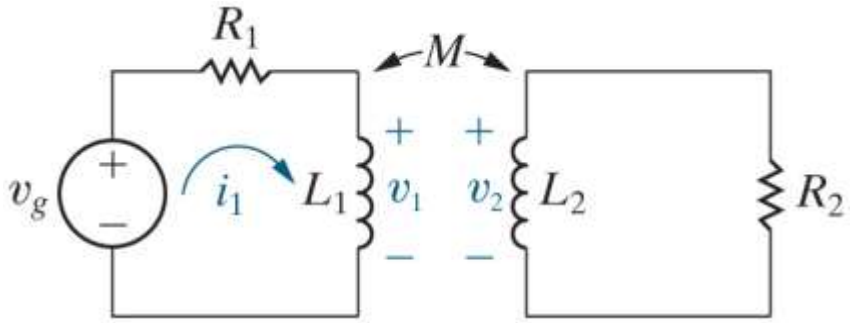
$$v_1(t) = L_1 \frac{di_1}{dt}$$

Bring in another coil, and continue moving it closer to the first coil – what happens? Eventually , the coils are close enough that the magnetic field surrounding the L_1 coil also envelops the L_2 coil, resulting in a voltage drop across the L_2 coil that is proportional to the first derivative of the i_1 current!

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (2 of 15)



$$v_1(t) = L_1 \frac{di_1}{dt}$$

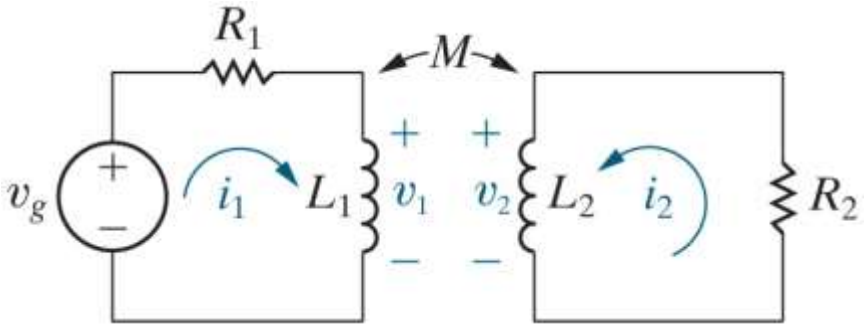
$$v_2(t) = M \frac{di_1}{dt}$$

But the voltage drop across the L_2 and the voltage drop across the R_2 resistor) creates current in the R_2-L_2 circuit. This current, labeled i_2, induces a time-varying magnetic field in the L_2 coil, causes additional voltage to drop across the L_2 coil, and, because the L_2 coil is close to the L_1 coil, causes additional voltage drop across the L_1 coil, proportional to the first derivative of the i_2, current!

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (3 of 15)



$$v_1(t) = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt}$$

$$v_2(t) = \pm M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

- We call L_1 and L_2 the **self-inductances** of the coils.
- We call M the **mutual inductance** of the coils.
- L_1 and L_2 and M are related by the following equation:

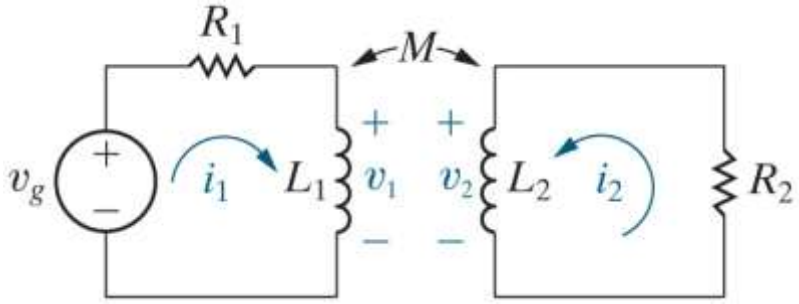
$$M = k\sqrt{L_1 L_2}, \qquad 0 \le k \le 1$$

- *k* is called the **coupling coefficient,** and is a measure of how much of the magnetic field of one coil envelops the other coil.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (4 of 15)



$$v_1(t) = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt}$$

$$v_2(t) = \pm M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

- The sign of the self-induced voltage for the two coils is determined by the passive sign convention.
- Based on the inductor voltage polarities and current directions shown in the circuit, the two self-induced voltage terms in the above equations are both positive.
- The sign of the mutually-induced voltage for the two coils is determined by physical examination of the coils, and is notated on the circuit schematic using dots.
- To determine the sign of the mutually-induced voltages for the two coils, use the **dot convention**.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (5 of 15)

The dot convention for mutually-coupled coils:

- When the current enters the dotted (un-dotted) terminal of one coil, the mutually-induced voltage drop is positive at the dotted (un-dotted) terminal of the other coil.

Dot Convention for Magnetically Coupled Inductor Coils

When the reference direction for a current enters the dotted terminal of a coil, the reference polarity of the voltage that it induces in the other coil is positive at its dotted terminal.

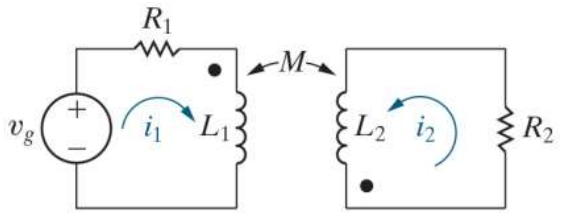
Alternate Dot Convention

When the reference direction for a current leaves the dotted terminal of a coil, the reference polarity of the voltage that it induces in the other coil is negative at its dotted terminal.

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (6 of 15)

The mesh current method (KVL) is always used to analyze circuits with mutually-coupled coils. Follow these steps:



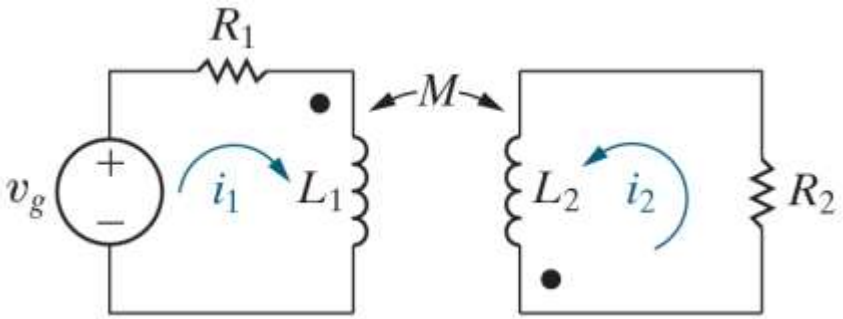
1. Draw the mesh currents and label them.
2. Pick a location on the left mesh and sum the voltage drops around the mesh in the direction of the mesh current, using the passive sign convention for the self-induced voltage terms.
3. Put the mutually-induced voltage term in the mesh equation, but don't include its sign yet.
4. Now look at the current in the right mesh – if it enters the dot (un-dot), the mutually-induced voltage term is positive at the dot (un-dot) of the other coil. Put a small + at the dot (un-dot).
5. If the current in the left mesh points at the + you just drew, the mutually-induced voltage term in the equation has a + sign; otherwise, it has a – sign.
6. Repeat steps 2 – 5 for the right mesh!

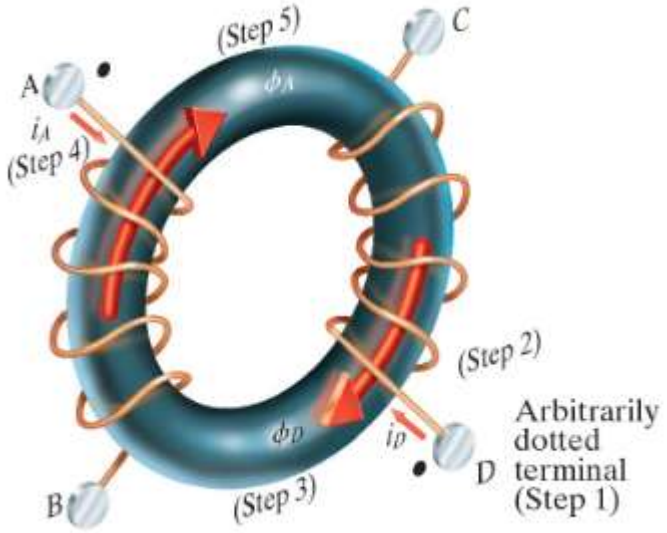
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (7 of 15)

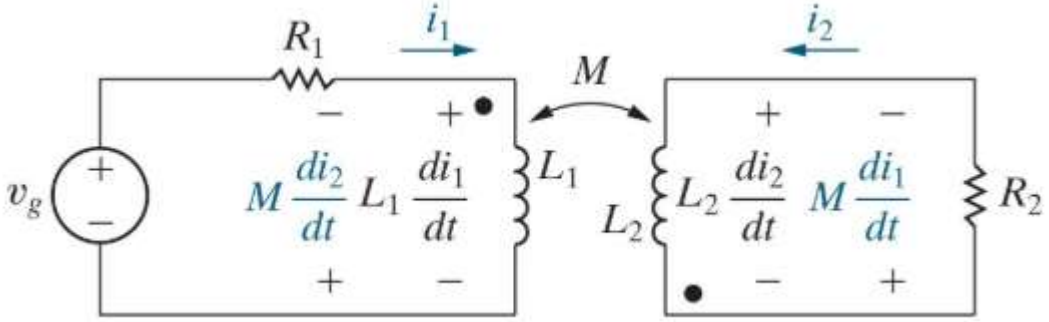
Write the mesh-current (KVL) equations for the two coils shown here:





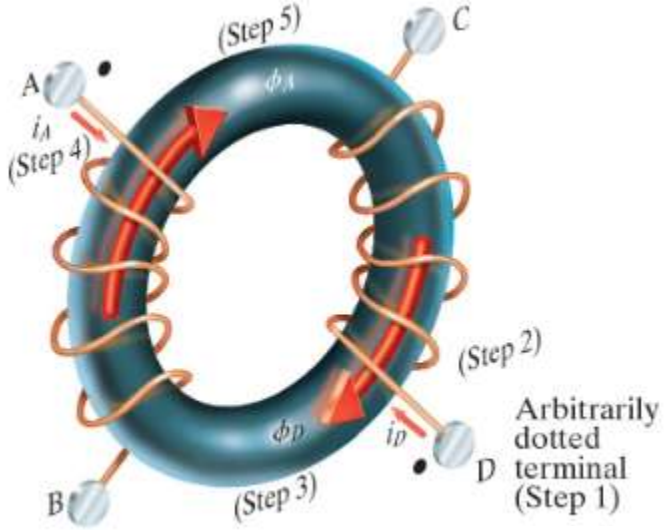
$$-v_g(t) + R_1 i_1 + L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} = 0$$

$$R_2 i_2 + L_2 \frac{di_2}{dt} - M \frac{di_1}{dt} = 0$$



Pearson

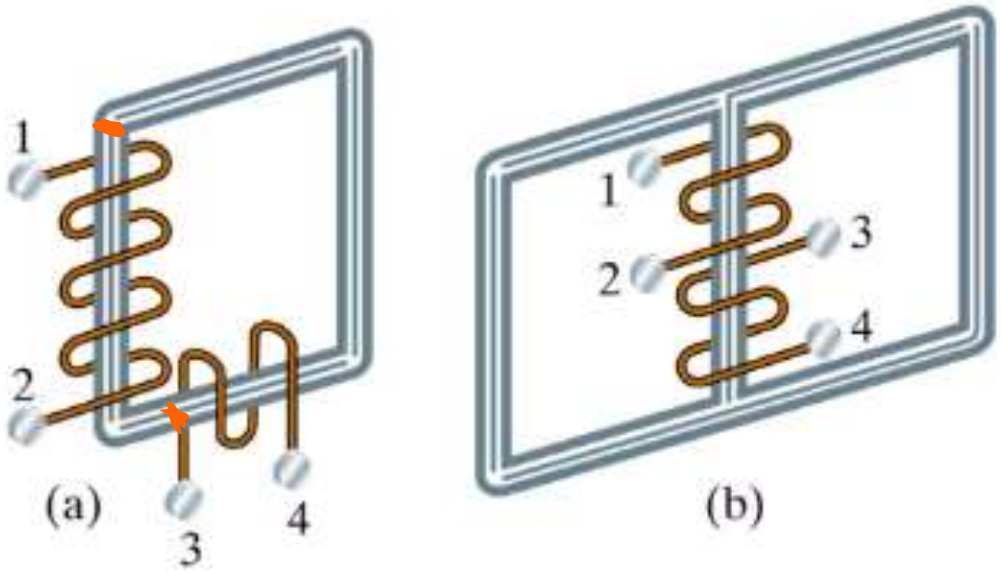
© 2024 Pearson Education Ltd. All Rights Reserved.



1. Arbitrarily select one terminal—say, the D terminal—of one coil and mark it with a dot.
2. Assign a current into the dotted terminal and label it i_D.
3. Use the right-hand rule[1] to determine the direction of the magnetic field established by i_D *inside* the coupled coils and label this field ϕ_D.
4. Arbitrarily pick one terminal of the second coil—say, terminal A—and assign a current into this terminal, showing the current as i_A.
5. Use the right-hand rule to determine the direction of the flux established by i_A *inside* the coupled coils and label this flux ϕ_A.
6. Compare the directions of the two fluxes ϕ_D and ϕ_A. If the fluxes have the same reference direction, place a dot on the terminal of the second coil where the test current (i_A) enters. If the fluxes have different reference directions, place a dot on the terminal of the second coil where the test current leaves. In Fig. 6.27, the fluxes ϕ_D and ϕ_A have the same reference direction, and therefore a dot goes on terminal A.

Pearson

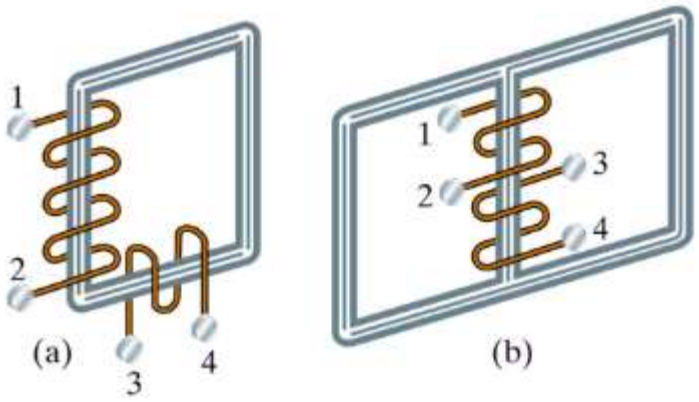
The physical construction of two pairs of magnetically coupled coils is shown in Fig. P6.44. Assume that the magnetic flux is confined to the core material in each structure. Show two possible locations for the dot markings on each pair of coils.



Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

The physical construction of two pairs of magnetically coupled coils is shown in Fig. P6.44. Assume that the magnetic flux is confined to the core material in each structure. Show two possible locations for the dot markings on each pair of coils.



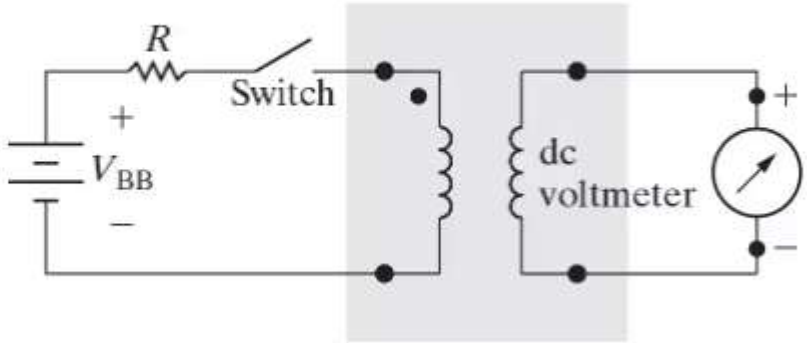
[a] Dot terminal 2; with current entering terminal 2, the flux is right-to-left coil 1-2. Assign the current into terminal 4; the flux is left-to-right in coil 3-4. The flux is in the same direction, due to the topology of the core, so dot terminal 4. Hence, 2 and 4 or 1 and 3.

[b] Dot terminal 1; with current entering terminal 1 the flux is up in coil 1-2. Assign the current into terminal 4; the flux is left-to-right in coil 3-4. Therefore the flux is in the same direction, due to the topology of the core, so dot terminal 4. Hence, 1 and 4 or 2 and 3.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

An experimental setup for determining polarity marks.



Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

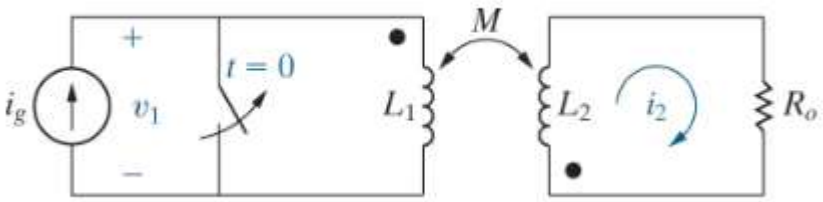
There is no energy stored in the circuit in Fig. P6.43 at the time the switch is opened.

a. Derive the differential equation that governs the behavior of i_2 if $L_1 = 10$ H, $L_2 = 40$ H, $M = 5$ H, and $R_o = 90\ \Omega$.

b. Show that when $i_g = 10e^{-t} - 10$ A, $t \geq 0$, the differential equation derived in (a) is satisfied when $i_2 = e^{-t} - 5e^{-2.25t}$ A, $t \geq 0$.

c. Find the expression for the voltage v_1 across the current source.

d. What is the initial value of v_1? Does this make sense in terms of known circuit behavior?



Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

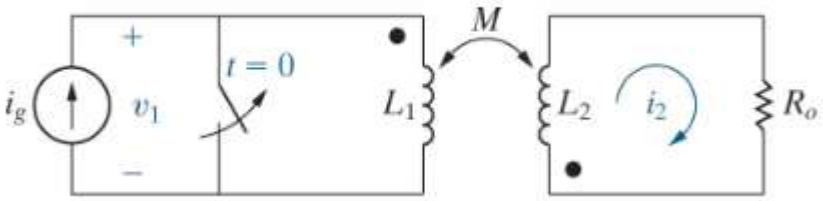
There is no energy stored in the circuit in Fig. P6.43 at the time the switch is opened.

a. Derive the differential equation that governs the behavior of i_2 if $L_1 = 10$ H, $L_2 = 40$ H, $M = 5$ H, and $R_o = 90\ \Omega$.

b. Show that when $i_g = 10e^{-t} - 10$ A, $t \geq 0$, the differential equation derived in (a) is satisfied when $i_2 = e^{-t} - 5e^{-2.25t}$ A, $t \geq 0$.

c. Find the expression for the voltage v_1 across the current source.

d. What is the initial value of v_1? Does this make sense in terms of known circuit behavior?



[a] $5\dfrac{di_g}{dt} + 40\dfrac{di_2}{dt} + 90i_2 = 0;$

$$40\frac{di_2}{dt} + 90i_2 = -5\frac{di_g}{dt}.$$

[b] $i_2 = e^{-t} - 5e^{-2.25t}$ A;

$$\frac{di_2}{dt} = -e^{-t} + 11.25e^{-2.25t}\ \text{A/s};$$

$$i_g = 10e^{-t} - 10\ A$$

$$\frac{di_g}{dt} = -10e^{-t}\ \text{A/s};$$

$$40[-e^{-t} + 11.25e^{-2.25t}] + 90[e^{-t} - 5e^{-2.25t}] = 50e^{-t};$$

$$-5[-10e^{-t}] = 50e^{-t}.$$

Pearson

eserved.

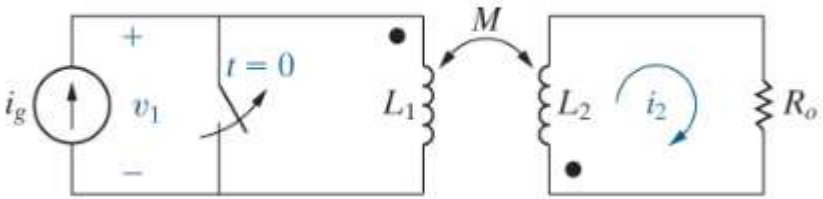
There is no energy stored in the circuit in Fig. P6.43 at the time the switch is opened.

a. Derive the differential equation that governs the behavior of i_2 if $L_1 = 10$ H, $L_2 = 40$ H, $M = 5$ H, and $R_o = 90\ \Omega$.

b. Show that when $i_g = 10e^{-t} - 10$ A, $t \geq 0$, the differential equation derived in (a) is satisfied when $i_2 = e^{-t} - 5e^{-2.25t}$ A, $t \geq 0$.

c. Find the expression for the voltage v_1 across the current source.

d. What is the initial value of v_1? Does this make sense in terms of known circuit behavior?



[c] $$v_1 = 10\frac{di_g}{dt} + 5\frac{di_2}{dt}$$

$$= 10(-10e^{-t}) + 5(-e^{-t} + 11.25e^{-2.25t})$$

$$= -105e^{-t} + 56.26e^{-2.25t}\ \text{V}, \qquad t > 0.$$

[d] $v_1(0) = -105 + 56.25 = -48.75$ V; Å

$$v_1(0) = 10\frac{di_g}{dt}(0) + 5\frac{di_2}{dt}(0)$$

$$= 10(-10) + 5(-1 + 11.25) = -100 + 56.26 = -48.75\ \text{V}.$$

Yes, the initial value of v_1 is consistent with known circuit behavior.

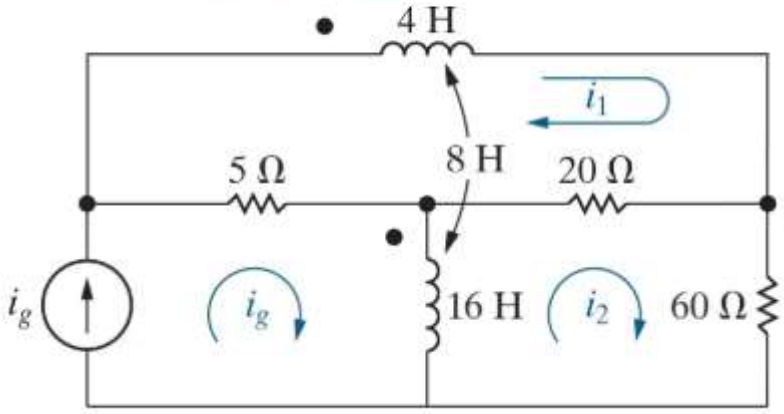
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Use the mesh-current method to write equations for the circuit in Fig. 6.29 in terms of the currents i_1 and i_2.

Figure 6.29

The circuit for Example 6.8.



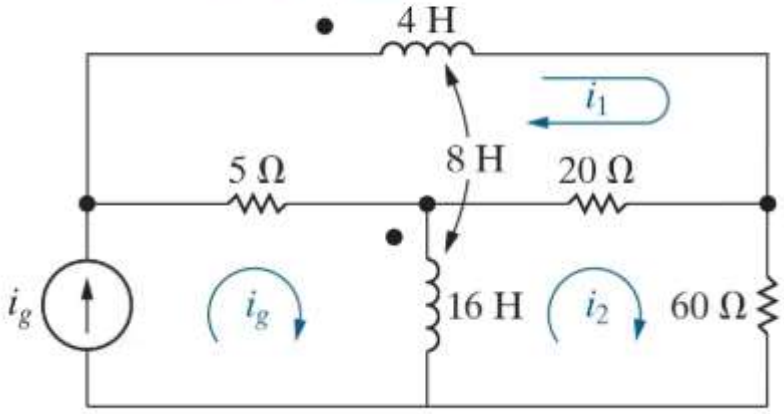
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Use the mesh-current method to write equations for the circuit in Fig. 6.29 in terms of the currents i_1 and i_2.

Figure 6.29

The circuit for Example 6.8.



$$4\frac{di_1}{dt} + 8\frac{d}{dt}\left(i_g - i_2\right) + 20(i_1 - i_2) + 5\left(i_1 - i_g\right) = 0.$$

$$20(i_2 - i_1) + 60i_2 + 16\frac{d}{dt}\left(i_2 - i_g\right) - 8\frac{di_1}{dt} = 0.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

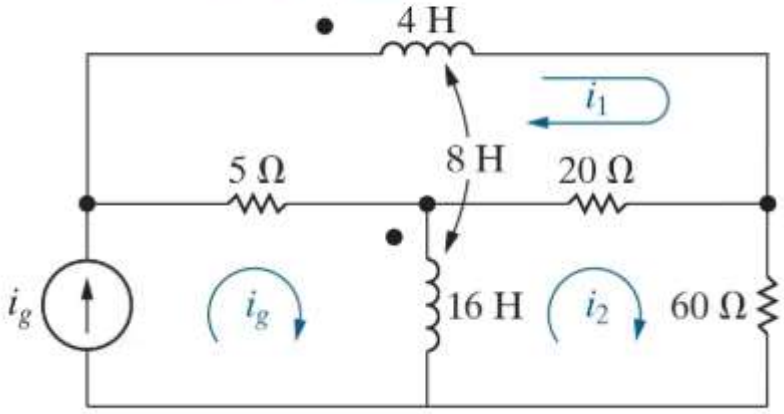
Verify that if there is no energy stored in the circuit at $t = 0$ and if $i_g = 16 - 16e^{-5t}$ A, the solutions for i_1 and i_2 are

$$i_1 = 4 + 64e^{-5t} - 68e^{-4t} \text{ A},$$

$$i_2 = 1 - 52e^{-5t} + 51e^{-4t} \text{ A}.$$

Figure 6.29

The circuit for Example 6.8.



Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

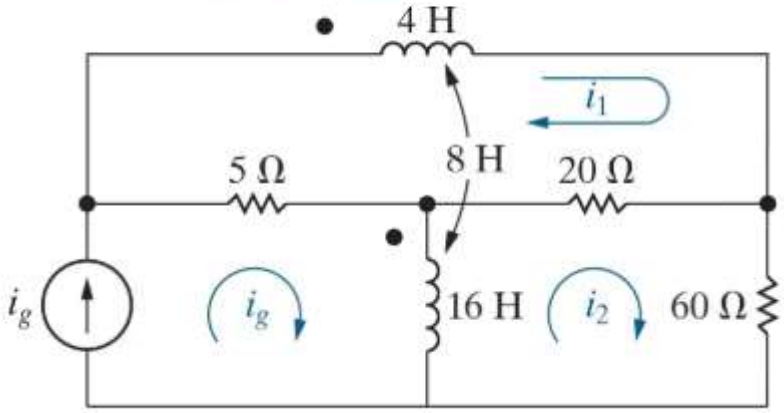
Verify that if there is no energy stored in the circuit at $t = 0$ and if $i_g = 16 - 16e^{-5t}$ A, the solutions for i_1 and i_2 are

$$i_1 = 4 + 64e^{-5t} - 68e^{-4t} \text{ A},$$

$$i_2 = 1 - 52e^{-5t} + 51e^{-4t} \text{ A}.$$

Figure 6.29

The circuit for Example 6.8.

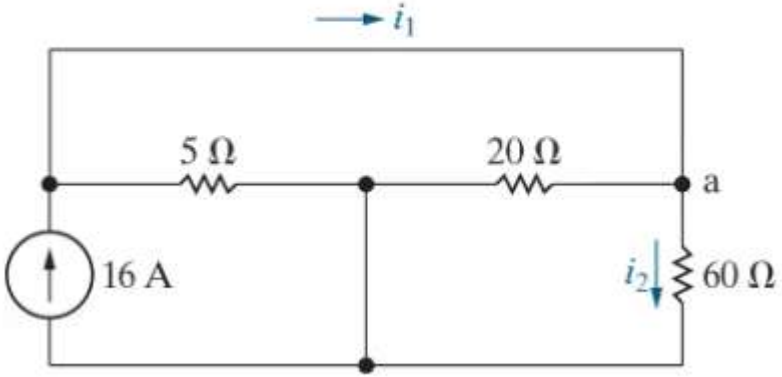


$$i_1(0) = 4 + 64 - 68 = 0,$$

$$i_2(0) = 1 - 52 + 51 = 0.$$

Figure 6.30

The circuit of Example 6.8 when $t = \infty$.



$$i_1(\infty) = \frac{60}{20} + \frac{60}{60} = 4 \text{ A}.$$

$$i_2(\infty) = \frac{60}{60} = 1 \text{ A}.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (8 of 15)

The two equations we just wrote are

A. A coupled system of linear algebraic equations

B. A coupled system of linear integral equations

C. A coupled system of linear differential equations

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (8 of 15)

The two equations we just wrote are

- ✗ A. A coupled system of linear algebraic equations
- ✗ B. A coupled system of linear integral equations
- ✓ C. A coupled system of linear differential equations

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (9 of 15)

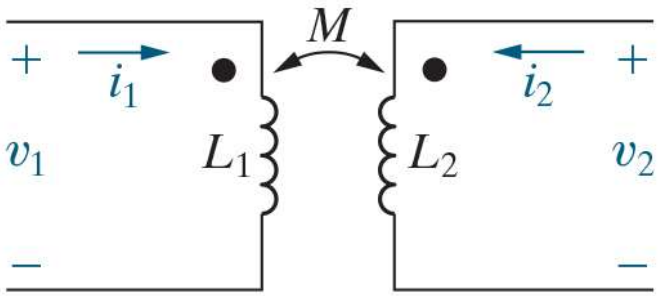
- We can’t do much in the time-domain with circuits that have mutually-coupled coils, since we can’t easily solve coupled systems of linear differential equations.
- We will revisit circuits with mutually-coupled coils once we examine the analysis of circuits with sinusoidal sources, starting in Chapter 9.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (10 of 15)

Use the circuit here to calculate the total energy stored in magnetically coupled coils.



Initially, assume both currents and the stored energy are zero. Then let i_1 increase from 0 to I_1 and calculate the stored energy:

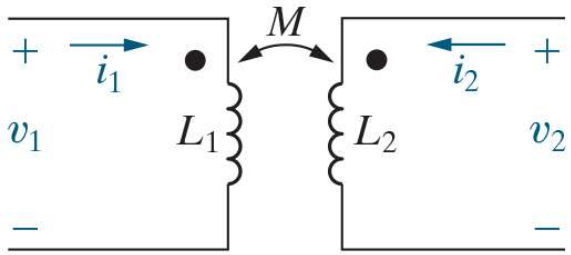
$$\int_0^{W_1} dw = L_1 \int_0^{I_1} i_1 di_1 \quad \Rightarrow \quad W_1 = \frac{1}{2} L_1 i_1^2.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (11 of 15)

Use the circuit here to calculate the total energy stored in magnetically coupled coils.



Now hold i_1 at I_1 and increase i_2 to I_2; during this increase, the voltage induced in coil 2 by i_1 is zero because i_1 is constant. The voltage induced in coil 1 by i_2 is Mdi_2/dt, so the power in the two coils is

$$p = I_1 M \frac{di_2}{dt} + i_2 v_2.$$

The total energy when i_2 **reaches** I_2 is

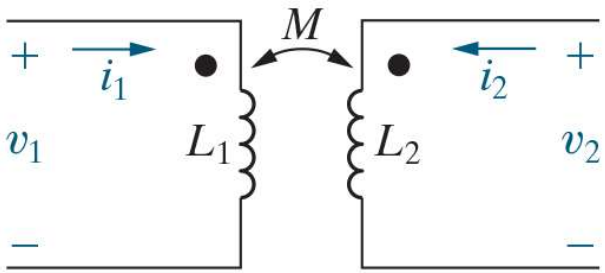
$$\int_{W_1}^{W} dw = \int_0^{I_2} I_1 M di_2 + \int_0^{I_2} L_2 i_2 di_2 \quad \Rightarrow \quad W = W_1 + I_1 I_2 M + \frac{1}{2} L_2 i_2^2.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (12 of 15)

Use the circuit here to calculate the total energy stored in magnetically coupled coils.



Note that in this example, both currents enter the dotted terminals. If one current enters a dotted terminal and the other current leaves, the algebraic sign of the term involving M reverses. Therefore, in general,

$$w(t) = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (13 of 15)

The energy in the magnetically coupled coils can never be negative. From the energy equation, the limiting value of M occurs when

$$\frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 - M i_1 i_2 = 0.$$

To find the limiting value of M, add and subtract $i_1 i_2 \sqrt{L_1 L_2}$ to both sides and simplify to get

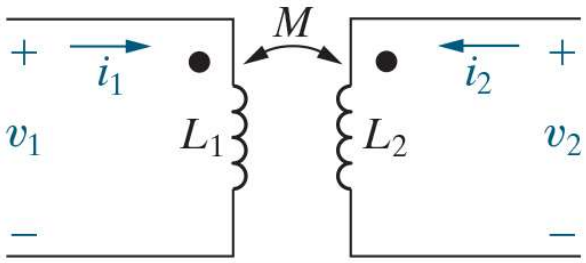
$$\left(\sqrt{\frac{L_1}{2}}i_1 - \sqrt{\frac{L_2}{2}}i_2\right)^2 + i_1 i_2\left(\sqrt{L_1 L_2} - M\right) = 0.$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (14 of 15)

Use the circuit here to calculate the total energy stored in magnetically coupled coils.



Therefore, the energy in the magnetically coupled coils is positive only if

$$\sqrt{L_1 L_2} \geq M.$$

Thus,

$$M = k\sqrt{L_1 L_2}, \qquad 0 \leq k \leq 1$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (15 of 15)

Suppose M = 40 mH, $L_1 = 25$ mH and $L_2 = 100$ mH in a pair of magnetically coupled coils. What is the coupling coefficient of the coils, k?

A. 0.8

B. 0.016

C. 1.25

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

Chapter 6 – Mutual Inductance (15 of 15)

Suppose M = 40 mH, $L_1 = 25$ mH and $L_2 = 100$ mH in a pair of magnetically coupled coils. What is the coupling coefficient of the coils, k?

- ✔ A. 0.8
- ✘ B. 0.016
- ✘ C. 1.25

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.